

Handbook

Setup Environment

1. Make a fresh installation of Manjaro.
2. Download the repo from Github.
3. a) Download models from the github links, and put them in a folder, say `/run/media/xxx/data/c491/c4-models-work/`.

```
Name
- ABL_a_asc_2x_lsct3sp_lsct3spva9r.zip
- baseline.zip
- DUAL_a_asc_0dancukmk7hnp_r45_C_trinorm_dsa3_va9r.zip
- DUAL_a_asc_0dancukmk7hnp_r45pt_C_trinorm_dsa3_va9r_lsct3sp_2x.zip
- DUAL_b_asc_0dancukmk7hnp_r45_C_trinorm_dsa3_va9r_lsct3sp_2x.zip
- DUAL_b_asc_0dancukmk7hnp_r45pt_C_trinorm_dsa3_va9r_lsct3sp_2x.zip
- DUAL_ch_asc_0dancukmk7hnp_r45_C_trinorm_dsa3_va9r_lsct3sp_2x.zip
```

- b) Run for i in \$(ls); do unzip \$i; done; to uncompress them
4. Download datasets from the github links and uncompress it to `/run/media/xxx/data/xxx/ssddata/`

```
> CUTE80      2 Files
> IC03_867    2 Files
> IC13_1015   2 Files
> IIIT5k_3000 2 Files
> SVT         2 Files
> dicts       29 Files
> mltrtrjp_hori 2 Files
> mltrkr_hori 3 Files
```

5. Uncompress the Athena.zip file to `/run/media/xxx/data/xxx/ssddata/`

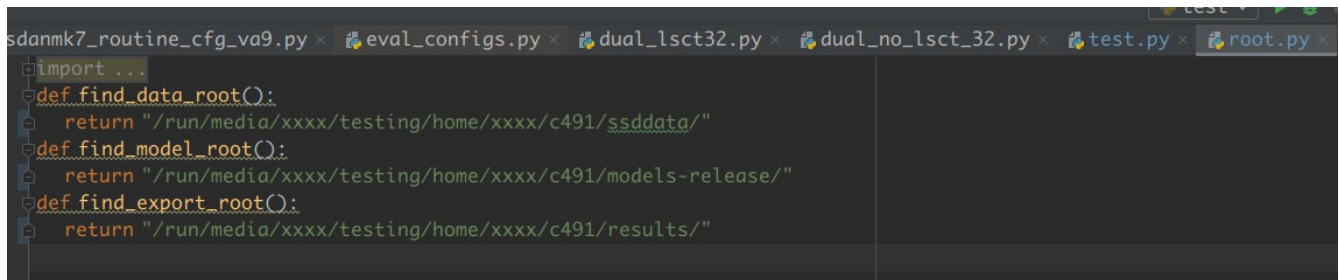
6. Install dependencies

- a) Install packages on paclist.txt in the git repo
- b) Install pip packages on the piplist.txt in the git repo
- c) Install pylcs-laser.zip

pylcs modded by a Github user, adds support to return the exact LCS

Source: <https://github.com/Meteorix/pylcs/issues/4>,

7. Change the paths in neko_sdk/root.py to the data and models uncompressed. If you do NOT want intermediate results, return None for find_export_root.



```
sdanmk7_routine_cfg_va9.py x eval_configs.py x dual_lsct32.py x dual_no_lsct_32.py x test.py x root.py x
import ...
def find_data_root():
    return "/run/media/xxxx/testing/home/xxxx/c491/ssddata/"
def find_model_root():
    return "/run/media/xxxx/testing/home/xxxx/c491/models-release/"
def find_export_root():
    return "/run/media/xxxx/testing/home/xxxx/c491/results/"
```

All done! You can now go to the next steps to check out if our framework lives up to its name.

Evaluation:

Going ~100 FPS on a laptop single batched is what we reported in the paper, now let's go 280+ FPS on that laptop going multi-batched. We actually liked the OSOCR way of testing on laptops so we also perform evaluations on laptops consuming around 230w (120w GPU+65W CPU + a little bit extra). This gives an insight into how the model may behave when deployed on a mobile platform that fits into vehicles or other power-consumption-sensitive computing platforms.

Reproducing Ablative Study

Base model: Run test_rej.py under DUAL_a_asc_Odancukmk7hnp_r45_C_trinorm_dsa3

```
GZSL
0.002199253616085609 4009 FPS: 454.69971843441704
0.0023821998219384194 4009
[base_chs_close_set_benchmark]test_accr
Accuracy: 0.378648, AR: 0.613850, CER: 0.386150, WER: 0.621352
[base_chs_close_set_benchmark]test_accr
Accuracy: 0.378648, AR: 0.613850, CER: 0.386150, WER: 0.621352
```

```
OSR
0.002173923619461345 4009 FPS: 459.9977621328665
0.0022628135471815004 4009
[base_chs_close_set_benchmark]test_accr
KACR: 0.734230, URCL: 0.215648, UPRE 0.987952, F 0.354021
[base_chs_close_set_benchmark]test_accr
```

73.42	21.56	98.79	35.40
-------	-------	-------	-------

```
GOSR
0.0021887277539872447 4009 FPS: 456.88642554026285
0.0023221606832396274 4009
[base_chs_close_set_benchmark]test_accr
KACR: 0.631792, URCL: 0.048260, UPRE 0.747826, F 0.090669
[base_chs_close_set_benchmark]test_accr
```

63.17	4.82	74.78	9.06
-------	------	-------	------

```
OSTR
0.0021791659736014805 4009 FPS: 458.8911593307014
0.0023068933303768305 4009
[base_chs_close_set_benchmark]test_accr
KACR: 0.668933, URCL: 0.461612, UPRE 0.961271, F 0.623711
[base_chs_close_set_benchmark]test_accr
```

66.89	46.16	96.12	62.37
-------	-------	-------	-------

CIL only: Run test_rej.py under DUAL_a_asc_Odancukmk7hnp_r45_C_trinorm_dsa3_va9r

```
GZSL
0.0021742276343718883 4009 FPS: 459.9334421986084
0.0023585923385905576 4009
[base_chs_close_set_benchmark]test_accr
Accuracy: 0.391868, AR: 0.608202, CER: 0.391798, WER: 0.608132
[base_chs_close_set_benchmark]test_accr
Accuracy: 0.391868, AR: 0.608202, CER: 0.391798, WER: 0.608132
```

```
OSR
0.0021648606804428806 4009 FPS: 461.92348959630186
0.0022521733084783328 4009
[base_chs_close_set_benchmark]test_accr
KACR: 0.765253, URCL: 0.650559, UPRE 0.969148, F 0.778521
[base_chs_close_set_benchmark]test_accr
```

76.52	65.05	96.91	77.85
-------	-------	-------	-------

```
GOSR
0.0021798392429289958 4009 FPS: 458.74942532749566
0.00231474205334902 4009
[base_chs_close_set_benchmark]test_accr
KACR: 0.673103, URCL: 0.524130, UPRE 0.846782, F 0.647487
[base_chs_close_set_benchmark]test_accr
```

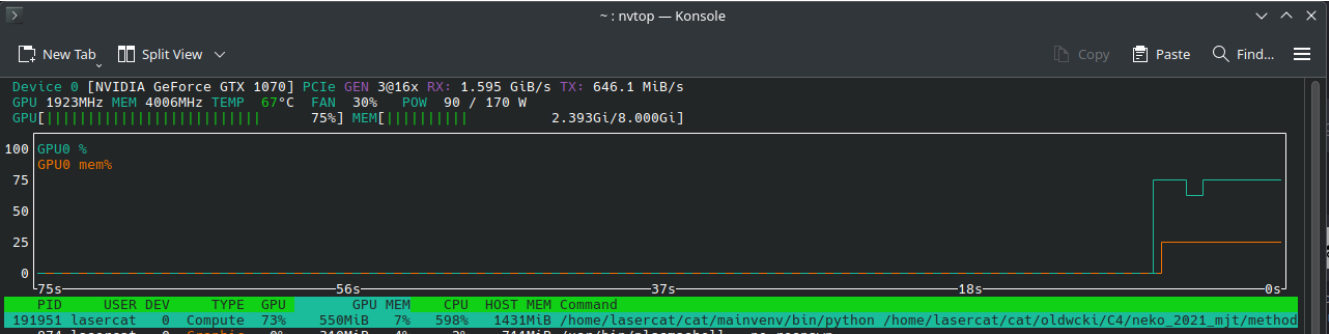
67.31	52.41	84.67	64.74
-------	-------	-------	-------

```
OSTR
0.00217630845943155 4009 FPS: 459.49368788521787
0.0023137217716969706 4009
[base_chs_close_set_benchmark]test_accr
KACR: 0.694561, URCL: 0.824511, UPRE 0.908088, F 0.864284
[base_chs_close_set_benchmark]test_accr
```

69.45	82.45	90.80	86.42
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ICL only and full model: Run test_rej.py under ABL_a_asc_2x_lsct3sp_lsct3spva9r

GZSL	OSR	GOSR	OSTR
0.0021965020194735186 4009 FPS: 455.26932874830266 0.002376268498170224 4009 [base_chs_lsctsp_2x_close_set_benchmark]test_accr Accuracy: 0.407333, AR: 0.612478, CER: 0.387522, WER: 0.592667 [base_chs_lsctsp_2x_close_set_benchmark]test_accr Accuracy: 0.407333, AR: 0.612478, CER: 0.387522, WER: 0.592667	OSR 0.002273420468028509 4009 FPS: 439.8658383097921 0.002368361553250182 4009 [base_chs_lsctsp_2x_close_set_benchmark]test_accr KACR: 0.742503, URCL: 0.630178, UPRE: 0.961867, F 0.761470 [base_chs_lsctsp_2x_close_set_benchmark]test_accr KACR: 0.742503, URCL: 0.630178, UPRE: 0.961867, F 0.761470	0.0022511610695773988 4009 FPS: 444.2152156565785 0.0023921580658209884 4009 [base_chs_lsctsp_2x_close_set_benchmark]test_accr KACR: 0.662775, URCL: 0.484287, UPRE: 0.829011, F 0.611406 [base_chs_lsctsp_2x_close_set_benchmark]test_accr KACR: 0.662775, URCL: 0.484287, UPRE: 0.829011, F 0.611406 JAP_lang ends	OSTR 0.00220222291798638 4009 FPS: 454.08534640424955 0.002330290881675745 4009 [base_chs_lsctsp_2x_close_set_benchmark]test_accr KACR: 0.698745, URCL: 0.740582, UPRE: 0.988719, F 0.816088 [base_chs_lsctsp_2x_close_set_benchmark]test_accr KACR: 0.698745, URCL: 0.740582, UPRE: 0.988719, F 0.816088
40.73	74.25 63.01 96.18 76.14	66.27 48.42 82.90 61.14	69.87 74.05 90.87 81.60
JAP_lang starts GZSL 0.002241421887689649 4009 FPS: 446.14537115578554 0.0023902302877835784 4009 [base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr Accuracy: 0.415505, AR: 0.617914, CER: 0.382086, WER: 0.584435 [base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr Accuracy: 0.415505, AR: 0.617914, CER: 0.382086, WER: 0.584435	OSR 0.002310443204431874 4009 FPS: 432.7799281524616 0.002397356856401557 4009 [base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr KACR: 0.750776, URCL: 0.775477, UPRE: 0.961288, F 0.858443 [base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr KACR: 0.750776, URCL: 0.775477, UPRE: 0.961288, F 0.858443	GOSR 0.002187244075585791 4009 FPS: 457.19634638040037 0.002328043164674526 4009 [base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr KACR: 0.676695, URCL: 0.687991, UPRE: 0.819519, F 0.748017 [base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr KACR: 0.676695, URCL: 0.687991, UPRE: 0.819519, F 0.748017	OSTR 0.0022848556986709103 4009 FPS: 437.66440068039975 0.0024160767291424193 4009 [base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr KACR: 0.718096, URCL: 0.863615, UPRE: 0.895205, F 0.879126 [base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr KACR: 0.718096, URCL: 0.863615, UPRE: 0.895205, F 0.879126
41.55 -	75.07 77.54 96.12 85.84	67.66 68.79 81.95 74.80	71.80 86.36 89.52 87.91



459 fps on 1070/12400f... This thing is darn CPU bounded. Or did torch get better again?

Xeons are super pricy in the nordic. Damn.

Reproducing Results On English, Japanese, and Korean

Regular model: Run test.py under ABL_a_asc_2x_lsct3sp_lsct3spva9r

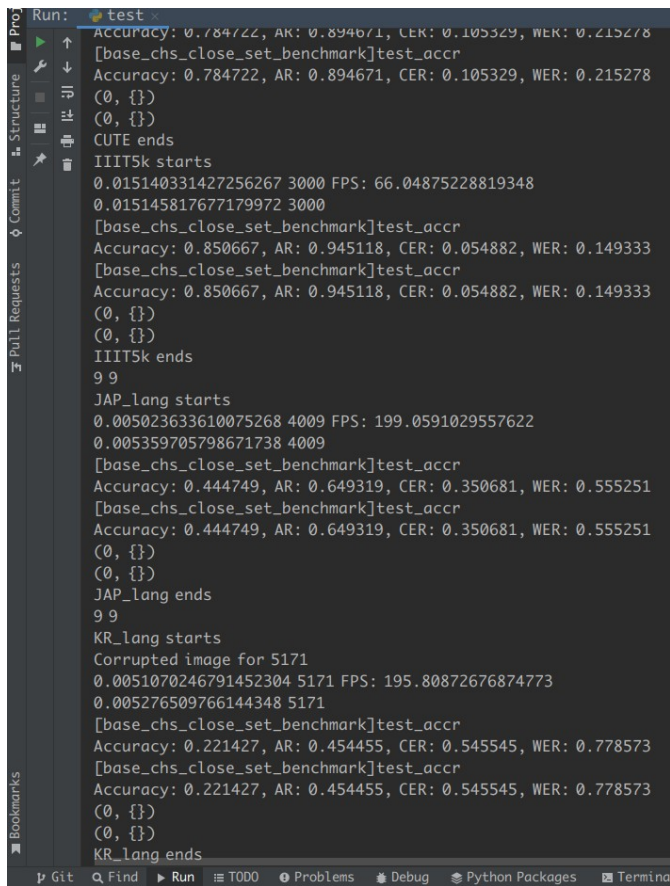
```
9 9
CUTE starts
/usr/lib/python3.10/site-packages/torch/utils/data/dataloader.py:
  create. Please be aware that excessive worker creation might get Data
  warnings.warn(_create_warning_msg(
0.010705261594719358 288 FPS: 93.41200970682262
0.014507010579109192 288
[base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr
Accuracy: 0.729167, AR: 0.870846, CER: 0.129154, WER: 0.270833
[base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr
Accuracy: 0.729167, AR: 0.870846, CER: 0.129154, WER: 0.270833
(0, {})
(0, {})
CUTE ends
IIIT5k starts
0.009958661953608195 3000 FPS: 100.41509639130614
0.009964030663172405 3000
[base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr
Accuracy: 0.832000, AR: 0.936669, CER: 0.063331, WER: 0.168000
[base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr
Accuracy: 0.832000, AR: 0.936669, CER: 0.063331, WER: 0.168000
(0, {})
(0, {})
IIIT5k ends
9 9
JAP_lang starts
0.003196817587426429 4009 FPS: 312.8110918599649
0.0033238392037981375 4009
[base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr
Accuracy: 0.415565, AR: 0.617967, CER: 0.382033, WER: 0.584435
[base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr
Accuracy: 0.415565, AR: 0.617967, CER: 0.382033, WER: 0.584435
(0, {})
(0, {})
JAP_lang ends
9 9
KR_lang starts
Corrupted image for 5171
0.0037015587281636723 5171 FPS: 270.15645932925554
0.0037829781426979526 5171
[base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr
Accuracy: 0.195320, AR: 0.438824, CER: 0.561176, WER: 0.804680
[base_chs_lsctsp_2x_va9r_close_set_benchmark]test_accr
Accuracy: 0.195320, AR: 0.438824, CER: 0.561176, WER: 0.804680
(0, {})
(0, {})
KR_lang ends
```

The FPSs are incoherent due to batch size.

Also note the running time includes prototype caching time, hence CUTE(288) has a lower FPS than IIIT5k(3000). In practical use, the FPS will be close to IIIT5k performance.

Large: Run test.py under

DUAL_a_asc_Odancukmk7hnp_r45pt_C_trinorm_dsa3_va9r_lsct3sp_2x



```
Run: test
Accuracy: 0.784722, AR: 0.894671, CER: 0.105329, WER: 0.215278
[base_chs_close_set_benchmark]test_accr
Accuracy: 0.784722, AR: 0.894671, CER: 0.105329, WER: 0.215278
(0, {})
(0, {})
CUTE ends
IIIT5k starts
0.015140331427256267 3000 FPS: 66.04875228819348
0.015145817677179972 3000
[base_chs_close_set_benchmark]test_accr
Accuracy: 0.850667, AR: 0.945118, CER: 0.054882, WER: 0.149333
[base_chs_close_set_benchmark]test_accr
Accuracy: 0.850667, AR: 0.945118, CER: 0.054882, WER: 0.149333
(0, {})
(0, {})
IIIT5k ends
9 9
JAP_lang starts
0.005023633610075268 4009 FPS: 199.0591029557622
0.005359705798671738 4009
[base_chs_close_set_benchmark]test_accr
Accuracy: 0.444749, AR: 0.649319, CER: 0.350681, WER: 0.555251
[base_chs_close_set_benchmark]test_accr
Accuracy: 0.444749, AR: 0.649319, CER: 0.350681, WER: 0.555251
(0, {})
(0, {})
JAP_lang ends
9 9
KR_lang starts
Corrupted image for 5171
0.0051070246791452304 5171 FPS: 195.80872676874773
0.005276509766144348 5171
[base_chs_close_set_benchmark]test_accr
Accuracy: 0.221427, AR: 0.454455, CER: 0.545545, WER: 0.778573
[base_chs_close_set_benchmark]test_accr
Accuracy: 0.221427, AR: 0.454455, CER: 0.545545, WER: 0.778573
(0, {})
(0, {})
KR_lang ends
```

Reproducing Close-set Benchmarks

Regular: run test.py under

DUAL_b_asc_Odancukmk7hnp_r45_C_trinorm_dsa3_va9r_lsct3sp_2x

```
Run: test (2) x
/usr/lib/python3.10/site-packages/torch/utils/data/dataloader.py:487
warnings.warn(_create_warning_msg(
9 9
CUTE starts
0.012077465653419495 288 FPS: 82.79882789125298
0.015858748720751867 288
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.795139, AR: 0.875235, CER: 0.124765, WER: 0.204861
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.795139, AR: 0.875235, CER: 0.124765, WER: 0.204861
(0, {})
(0, {})
CUTE ends
IIIT5k starts
0.010423674662907919 3000 FPS: 95.93545772859218
0.010429205020268758 3000
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.892333, AR: 0.955465, CER: 0.044535, WER: 0.107667
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.892333, AR: 0.955465, CER: 0.044535, WER: 0.107667
(0, {})
(0, {})
IIIT5k ends
SVT starts
0.010439500558136788 647 FPS: 95.79002313674641
(0, {})
IIIT5k ends
SVT starts
0.010439500558136788 647 FPS: 95.79002313674641
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.822257, AR: 0.928346, CER: 0.071654, WER: 0.177743
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.822257, AR: 0.928346, CER: 0.071654, WER: 0.177743
(0, {})
(0, {})
SVT ends
IC03 starts
0.010752115832618089 867 FPS: 93.00495042718535
0.010770864552692558 867
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.925029, AR: 0.967801, CER: 0.032199, WER: 0.074971
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.925029, AR: 0.967801, CER: 0.032199, WER: 0.074971
(0, {})
(0, {})
IC03 ends
IC13 starts
0.010621431425874456 1015 FPS: 94.14926857823879
0.010637677479260074 1015
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.903448, AR: 0.967724, CER: 0.032276, WER: 0.096552
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.903448, AR: 0.967724, CER: 0.032276, WER: 0.096552
```

Large: run test.py under

DUAL_b_asc_Odancukmk7hnp_r45pt_C_trinorm_dsa3_va9r_lsct3sp_2x

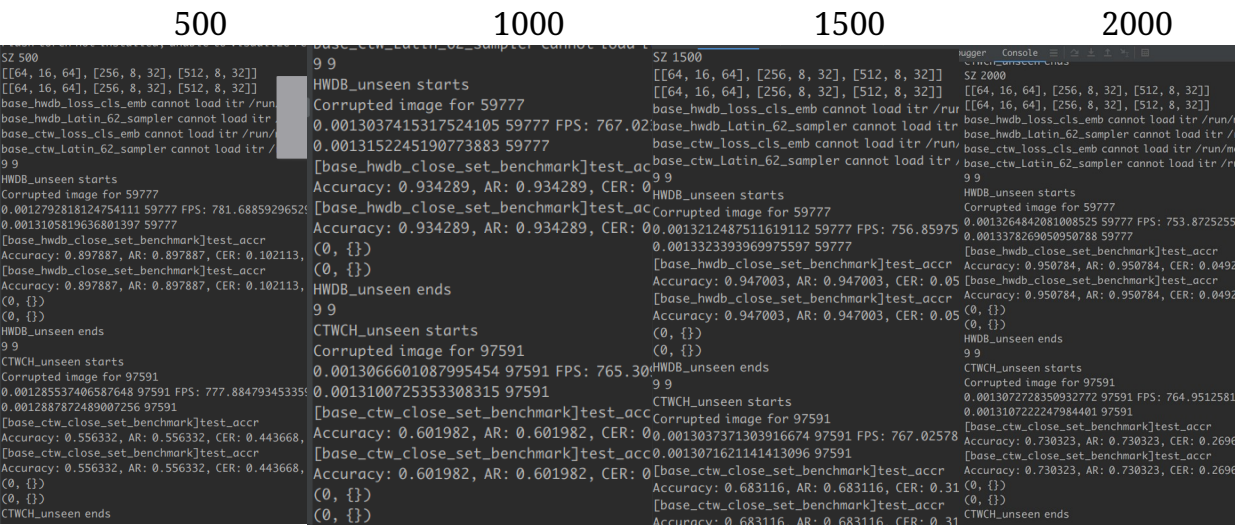
```
/usr/lib/python3.10/site-packages/torch/utils/data/dataloader.py:487
warnings.warn(_create_warning_msg(
9 9
CUTE starts
0.01571953876150979 288 FPS: 63.61509807454137
0.019835591316223145 288
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.815972, AR: 0.900313, CER: 0.099687, WER: 0.184028
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.815972, AR: 0.900313, CER: 0.099687, WER: 0.184028
(0, {})
(0, {})
CUTE ends
IIIT5k starts
0.014877764304478963 3000 FPS: 67.2143999282842
0.014882989486058554 3000
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.915333, AR: 0.965486, CER: 0.034514, WER: 0.084667
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.915333, AR: 0.965486, CER: 0.034514, WER: 0.084667
(0, {})
(0, {})
IIIT5k ends
SVT starts
0.015122226069478384 647 FPS: 66.12783034756558
(0, {})
IIIT5k ends
SVT starts
0.015122226069478384 647 FPS: 66.12783034756558
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.842349, AR: 0.945205, CER: 0.054795, WER: 0.157651
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.842349, AR: 0.945205, CER: 0.054795, WER: 0.157651
(0, {})
(0, {})
SVT ends
IC03 starts
0.015496390363611839 867 FPS: 64.53115703307076
0.015515541397851506 867
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.922722, AR: 0.967801, CER: 0.032199, WER: 0.077278
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.922722, AR: 0.967801, CER: 0.032199, WER: 0.077278
(0, {})
(0, {})
IC03 ends
IC13 starts
0.017553278025735187 1015 FPS: 56.9694161132685
0.017570234636955073 1015
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.903448, AR: 0.969950, CER: 0.030050, WER: 0.096552
[base_mjst_close_set_benchmark]test_accr
Accuracy: 0.903448, AR: 0.969950, CER: 0.030050, WER: 0.096552
(0, {})
(0, {})
IC13 ends
```


Reproducing Zero-shot Character Benchmarks

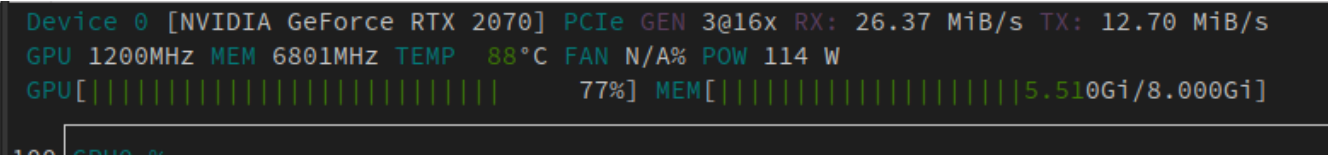
Run test.py under

DUAL_ch_asc_Odancukmk7hnp_r45_C_trinorm_dsa3_va9r_lsct3sp_2x

Note the datasets are not uploaded for review due to the size limitation.



Speed drops after the card overheats, poor little card.



Recognition of some Greek Family Languages

Run test_athena.py under

Name	Modified
lang_Glagolitic	3 minute
results	3 minute
lang_Georgian	3 minute
results	3 minute
lang_Armenian	3 minute
results	3 minute
16.txt	3 minute
16.jpg	3 minute
15.txt	3 minute
15.jpg	3 minute
14.txt	3 minute

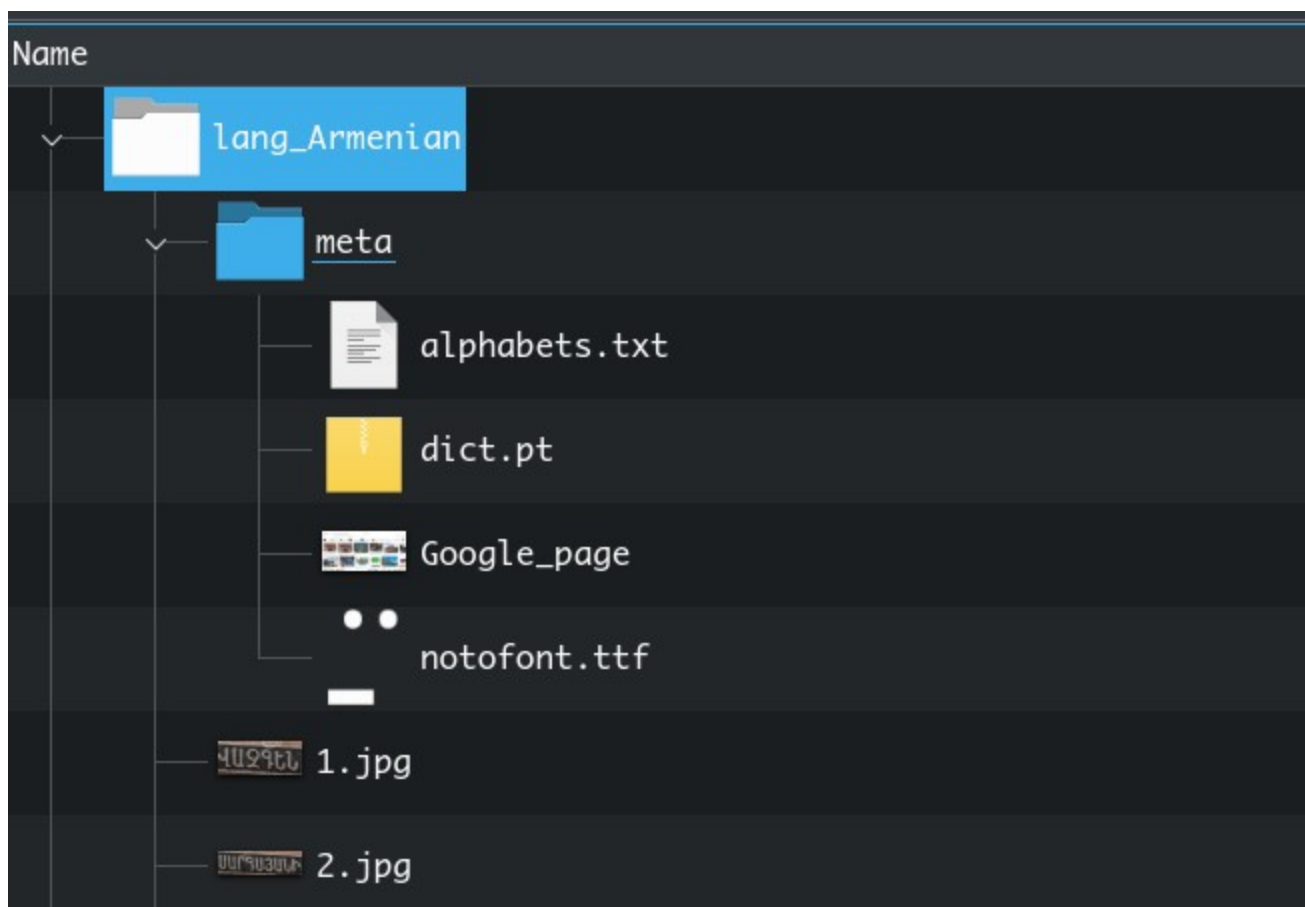
DUAL_a_asc_Odancukmk7hnp_r45pt_C_trinorm_dsa3_va9r_lsct3sp_2x

If you have your own language in mind, say xxxx language, you can test it with the following steps:

- a) add a folder named lang_xxxx in the Athena folder (the lang_xxxx part is mandatory for the testing script to pick it up)
- b) add the images in question to the folder
- c) Add a folder name meta (again it has to be named as `meta`)
- d) Download Noto sans regular font in question to the meta folder and name it as `notofont.ttf`
- e) add the alphabets file, each character per line
if the character has more than one possible shape, write all of them in one line

	alphabets.txt
1	Աա
2	Բբ
3	Գգ
4	Դդ
5	Եե
6	Զզ
7	Էէ
8	Ըը
9	Թթ
10	Ճճ
11	Իի
12	Լլ
13	Խյ
14	Օօ

f) Done! Don't worry about the dict file, the framework will build it from notofont and alphabets



System information

Program 10 Super I/O

Class	Description
=====	
system	Computer
bus	Motherboard
memory	24GiB System memory
processor	Intel(R) Core(TM) i5-9400 CPU @ 2.90GHz
bridge	8th Gen Core Processor Host Bridge/DRAM Registers
bridge	6th-10th Gen Core Processor PCIe Controller (x16)
display	TU106M [GeForce RTX 2070 Mobile]
multimedia	TU106 High Definition Audio Controller
display	CoffeeLake-S GT2 [UHD Graphics 630]